

ARTURO ARIAS

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EDUCATION

Carnegie Mellon University, Heinz College, Pittsburgh, PA **Expected 12/2026**
Master of Information Systems Management – Business Intelligence and Data Analytics (MISM-BIDA)
Coursework: Fundamentals of Operationalizing AI, Machine Learning, Artificial Intelligence, Statistics, Econometrics

Universidad de Ingenieria y Tecnologia – UTEC, Lima, Peru **04/2015 – 08/2020**
Bachelor of Chemical Engineering
Coursework: Process Control, Modeling and Simulation, Software Design, Programming, Chemical Process Analysis

SKILLS AND CERTIFICATIONS

Machine Learning and Artificial Intelligence: Deep Learning, Neural Networks, Natural Language Processing (NLP), Computer Vision, Retrieval-Augmented Generation (RAG), Fine-tuning, Prompt Engineering, Classification, OCR

Programming and Frameworks: Python, SQL, Java, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenAI API

Data Engineering: ETL/ELT Pipelines, Data Processing, Data Modeling, CI/CD Pipelines, Model Deployment, Docker

Cloud and Scalable AI Platforms: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)

Data Science and Statistical Methods: Data Analytics, Statistical Analysis, Hypothesis Testing, Regression, Time Series, Causal Inference, Experimental Design (A/B Testing), Data Visualization, Data Storytelling

Core Professional Skills: Analytical Problem-Solving, Cross-functional Collaboration, Learning Agility, Attention to Detail

Certifications: Certified Associate Python Programmer, Designing and Building AI Products and Services (MIT xPRO)

Languages: Spanish (native), English (professional), German (A2), Italian (basic)

WORK EXPERIENCE

Independent / Incarail, Lima, Peru **05/2025 – 08/2025**
Business Intelligence Specialist Consultant

- **Data Engineering | Analytics Systems:** Engineered Power BI dashboards on Microsoft Fabric, delivering e-commerce sales and web performance insights
- **Cloud Data Engineering | Performance Optimization:** Streamlined Microsoft Fabric dashboards; reduced file size by 92% and the refresh processing time by 90%, improving the user experience (UX) and lowering capacity costs
- **Python (PySpark) | Distributed Data Processing:** Automated Optical Character Recognition (OCR) workflows for PDFs in Fabric using PySpark in Python, eliminating manual effort and accelerating reporting
- **SQL | Query Optimization:** Optimized BigQuery queries and partitions, reducing execution time by 80%+ and lowering compute spend

Oncosalud, Lima, Peru **11/2024 – 05/2025**
Data Analyst

- **Data Engineering | Python | SQL:** Centralized and integrated Google Analytics (GA4), leads, and affiliation data in Amazon Web Services (AWS), enabling scalable analytics and data-driven decision-making; automated multi-source reporting, reducing data processing time by 95%
- **Data Engineering | Analytics Instrumentation:** Administered Google Analytics (GA4) and Google Tag Manager (GTM), closing tracking gaps and ensuring accurate, governed data collection

Alicorp, Lima, Peru **05/2021 – 03/2024**
Researcher

- **Data Science | Python:** Spearheaded nutritional analytics workflows in Python, introducing comparative indices to assess and benchmark performance
- **Data Visualization | Data Storytelling:** Produced executive-level visualizations that translated analytical findings into actionable business recommendations

PROJECTS

Real-Time Crypto Volatility Detection AI Service

- **Machine Learning | MLOps | Real-Time Data Streaming:** Built an end-to-end Coinbase volatility monitoring pipeline using Kafka, windowed feature engineering, and XGBoost to predict 60-second volatility spikes; deployed FastAPI inference using Docker for containerization, with MLflow tracking, Evidently drift reports, Prometheus metrics, Grafana dashboards, and baseline rollback support.